

MATLAB

matlab.mathworks.com

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Copy FigurePrintShow CodeFILE

TitleSubtitleX-LabelY-LabelZ-LabelLegendColorbarGridX-GridY-Grid

LABELS AND ANNOTATIONS

MATLAB Drive

Files

Name	Value	Size	Class
PG_dB	[8.4510,11.76...	1×8	double
PG_linear	[7,15,31,63,1...	1×8	double
PN_length	[7,15,31,63,1...	1×8	double
T	8×2 table	8×2	table

untitled * X

```
1 clc;
2 clear;
3 close all;
4 % PN sequence lengths
5 PN_length = [7 15 31 63 127 255 511 1023];
6 % Processing Gain (Linear)
7 PG_linear = PN_length;
8 % Processing Gain in dB
9 PG_dB = 10*log10(PG_linear);
10 % Display table
11 T = table(PN_length',PG_dB',...
12         'VariableNames',{'PN_Length','Processing_Gain_dB'});
13 disp('Processing Gain Table');
14 disp(T);
15 % Plot
16 figure;
17 stem(PN_length,PG_dB,'filled');
18 grid on;
19 xlabel('PN Sequence Length');
20 ylabel('Processing Gain (dB)');
21 title('Spread Spectrum: Processing Gain vs PN Sequence Length');
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

255	24.003
511	27.084
1023	30.099

>>

Figure 1 X

Spread Spectrum: Processing Gain vs PN Sequence Length...

PN Sequence Length	Processing Gain (dB)
7	8.4510
15	11.7627
31	14.9133
63	17.7942
127	20.5428
255	23.2832
511	27.0838
1023	30.0992

32°C Mostly sunny

Search

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